

A Wireless Stress Mapping System for Orthodontic Brackets Using CMOS Integrated Sensors

Matthias Kuhl, *Student Member, IEEE*, Pascal Gieschke, Daniel Rossbach, Sascha A. Hilzensauer, Thanapon Panchaphongsaphak, Patrick Ruther, *Member, IEEE*, Bernd G. Lapatki, Oliver Paul, *Member, IEEE*, and Yiannos Manoli, *Senior Member, IEEE*

Abstract—A wireless stress mapping system for the development of “smart brackets” is presented. The system is designed for monitoring the forces and moments applied to individual teeth during orthodontic therapy which may contribute to improve treatment effectiveness. It comprises a stress mapping chip fabricated in a $0.35\text{-}\mu\text{m}$ process and a micro coil produced by gold electroplating in a photoresist mask. Twenty-four transistor-based stress sensors for measuring either in-plane shear stress or the difference of in-plane normal stresses are strategically distributed over the chip area. The sensor signals are processed by a variable-gain differential difference amplifier and digitized by a 10-bit SAR ADC, enabling a resolution down to 11 kPa at highest gain. A wireless interface conditions the energy received at 13.56 MHz and transmits the data by load modulation. With dimensions of $2 \times 2.5 \times 0.73 \text{ mm}^3$ and a power consumption of 1.75 mW, the system lays the foundation for the assembly of smart brackets as innovative tools in orthodontic therapy.

Index Terms—Automatic gain tuning, biomedical devices, CMOS integrated circuits, CMOS MEMS, differential difference amplifier, load modulation, microcoil, orthodontic bracket, piezo FET, RFID, stress mapping, stress sensor, variable gain, wireless.

I. INTRODUCTION

WITH ongoing miniaturization in technology and increasing complexity in assembly and packaging, stress sensing microsystems are becoming essential for the development and evaluation of IC packages [1], [2]. Additionally, integrated stress sensor systems are attractive for miniaturized force and torque sensing, e.g., in coordinate measurement machines [3] and in various medical applications, such as osteosynthesis [4] or orthodontic therapy [5]. Fixed orthodontic appliances consist of multiple brackets bonded to individual teeth and an archwire inserted in the bracket slots. Such appliances exert a 3-D force-moment system on each tooth resulting

in mechanical loading of the soft tissues between the root and the alveolar bone (i.e., the periodontium). This load stimulates alveolar bone remodelling and thus induces tooth movement [6]. With current appliances, the effectiveness and the outcome of the orthodontic treatment depends to a great extent on the skill and the experience of the clinician. The presented system is designed for the next generation of fixed orthodontic appliances consisting of “smart brackets,” which provide the orthodontist with objective feedback on the forces and moments applied to each individual tooth during therapy [5]. This should allow for optimization of the applied mechanical stimulus. An accurate 3-D control of the induced translations and rotations of teeth is important for the reduction of negative side effects such as pain and irreversible root resorption due to overloading of teeth [7].

Fig. 1(a) presents the smart bracket in its target application, i.e., encapsulation into the base of an orthodontic ceramic bracket. Therefore, the chip dimensions are restricted to $2 \times 2.5 \times 1 \text{ mm}^3$. For mechanical stimulation of the periodontal tissues, the orthodontist has to exert a suitable set of force and moment components on each individual tooth. Typical forces and moments are in the range of $\pm 1 \text{ N}$ and $\pm 1 \text{ Ncm}$ and usually do not exceed $\pm 5 \text{ N}$ and $\pm 5 \text{ Ncm}$, respectively [8], whereby a resolution of 10% of the nominal values is considered sufficient for orthodontic precision.

The desired mechanical load is usually generated by elastic deformation of an archwire inserted in the bracket slot (Fig. 1(a)). The applied forces and moments cause mechanical stresses in the bracket, which are transferred into its base and thus result in a complex stress distribution on the sensor chip surface. Finite element (FE) simulations with a realistic smart bracket model indicate that measurements of shear stress σ_{xy} and the difference of normal stresses ($\sigma_{xx} - \sigma_{yy}$) are sufficient for reliable determination of the 3-D force-moment system exerted on the bracket and tooth [8] [Fig. 1(c)].

Theoretically, a total of six stress sensors are sufficient for this determination. However, nonlinearities, temperature variations, and variability in the system assembly make a certain redundancy desirable. FE simulations revealed a high sensing accuracy with 10 shear and 14 normal stress sensors placed at designated spots of maximum stress density [Fig. 1(b)]. A mean error below 0.12 N and 0.04 Ncm was reported, after calibration of the assembled system [8]. The calibration also compensates for misalignments, which might occur during assembly and packaging of the system. According to these results, nominal loads of 1 N or 1 Ncm on the bracket generate stress values from 0.06 to 8.18 MPa at the single sensors. The required resolution is then derived from the least noticeable out of the six separate load

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M. Kuhl, P. Gieschke, S. A. Hilzensauer, P. Ruther, O. Paul, and Y. Manoli are with the Department of Microsystems Engineering (IMTEK), University of Freiburg, 79110 Freiburg, Germany (e-mail: kuhl@imtek.de).

D. Rossbach and Y. Manoli are with the HSG-IMIT - Institute of Micromachining and Information Technology, 78052 Villingen-Schwenningen, Germany (e-mail: daniel.rossbach@hsg-imit.de).

T. Panchaphongsaphak and B. G. Lapatki are with the Department of Orthodontics, University of Ulm, 89081 Ulm, Germany (e-mail: bernd.lapatki@uniklinik-ulm.de).

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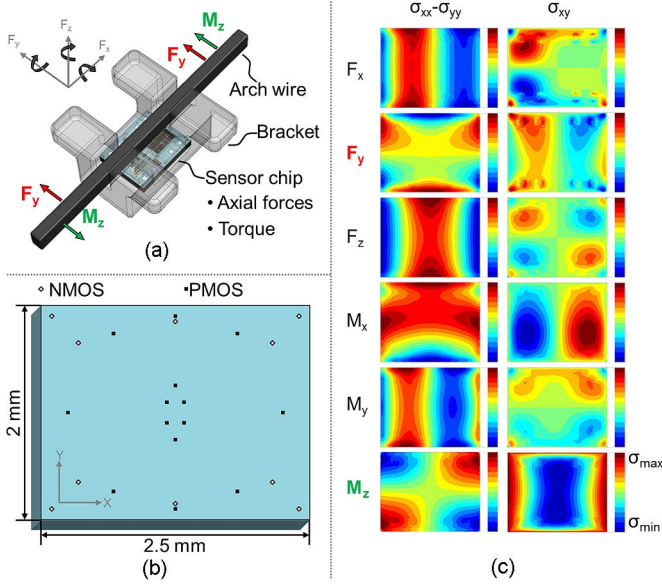


Fig. 1. Conceptual design of a smart bracket for measuring the forces and moments applied to individual teeth for therapeutic tooth movement using fixed orthodontic appliances. (a) Illustration of two exemplary stress situations on the bracket. (b) Positions of the 24 sensors on the chip surface. (c) Resulting shear and normal stress distribution according to FE simulation (based on data from [8]). Each color bar is normalized to the minimum (blue) and maximum (red) stress level of the respective stress map.

cases. Simulations show that applying a force $F_y = 1$ N results in an average stress value of 0.5 MPa for all 24 sensors. Adding a factor of 2 as safety margin and aiming for a resolution of 10% reveals a required resolution of 25 kPa.

In [9] and [10], a CMOS stress mapping chip was presented, proving the concept of distributed stress sensing with digital output on a chip sufficiently small for integration in an orthodontic bracket. The performance in terms of stress resolution was in good agreement with the smart bracket requirements; except for the high-power consumption (20 mW) and the wire-bonded connections, which are not compatible with clinical applications. In order to enable a simple and efficient handling of the appliance in clinical practice and to obtain a fully encapsulated, biocompatible measurement device, wireless communication is mandatory. A first wireless stress mapping chip was presented in [11]. This paper now presents an assembled prototype with a revised IC, resulting in enhanced stress-sensing capability and improved efficiency.

An overview of the sensor system is given in Section II. The various system components are described and characterized in Section III, while the measurements of the assembled system are presented in Section IV. Section V concludes the results of the paper.

II. SYSTEM OVERVIEW

Fig. 2 gives an overview of the system components. The 24 sensors, 10 sensitive to σ_{xy} and 14 to $(\sigma_{xx} - \sigma_{yy})$, can be consecutively connected to the readout unit by a 5-bit multiplexer (MUX). Furthermore, the processed sensor can be biased in four directions by an additional MUX, which enables an offset-compensated operation due to the Wheatstone bridge architecture of the sensors. A variable-gain differential difference amplifier

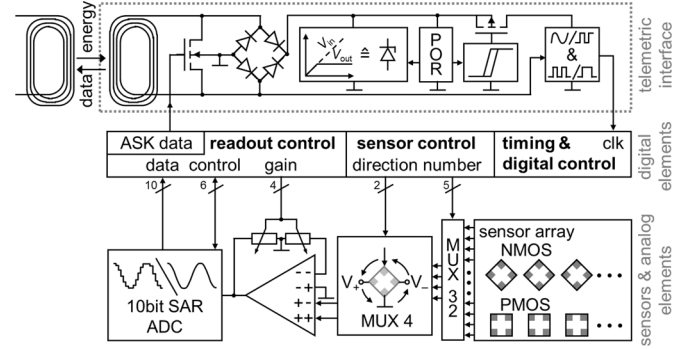


Fig. 2. Schematic overview of the smart bracket system.

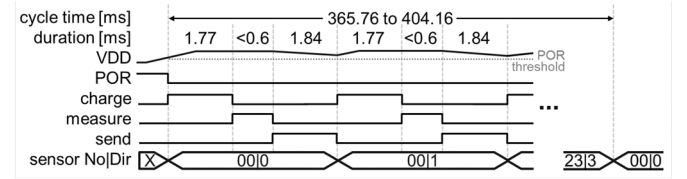


Fig. 3. Timing diagram of the cyclic data acquisition.

(DDA) together with a 10-bit successive approximation (SAR) analog-to-digital converter (ADC) performs the sensor readout. The system is powered wirelessly at 13.56 MHz with the sensor data being transmitted to an external unit by amplitude shift keying (ASK) modulation at 847.5 kHz, which is a submultiple of the carrier frequency. For inductive coupling, a 35-winding micro coil with gold leads has been developed, which outperforms a CMOS integrated coil in terms of the quality factor (see below). The assembly of chip and coil occupies a volume of $2 \times 2.5 \times 0.73$ mm³, connected via flip-chip bonding. The circuit is implemented in the 0.35 μ m CMOS process XH035 [12], and the coil is fabricated in-house using gold (Au) electroplating in a thick photoresist mask.

Fig. 3 illustrates the operation of the system. After onset of wireless energy transmission, rectification, and voltage stabilization, the system is activated by a power-on-reset (POR) circuit attesting a suitable supply voltage. The stress measurement itself is organized as a cyclic data acquisition task alternating between three states: 1) reading out one sensor in one direction; 2) sending the sampled data by load modulation; and 3) recharging the buffer capacitor. In each cycle, another sensor direction is selected. After processing of all 24 sensors in four directions each, the cycle is restarted from the beginning.

III. SYSTEM COMPONENTS

A. External Unit and Wireless Interface

The external power transmission and data reception unit is designed for 13.56 MHz, in agreement with ISM band definition for medical RFID reader systems. The external unit consists of a bipolar transistor driving a parallel RLC resonant circuit. To extract the modulated sensor data, a capacitively coupled filter is utilized, mainly based on rectification and differentiation of the 847.5-kHz-modulated excitation voltage. At this prototype stage, the data processing is performed on a personal computer.

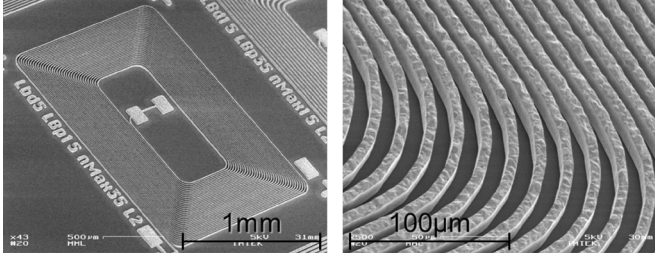


Fig. 7. Micrograph of the coil produced by gold electroplating.

phase for the supply voltage buffer. Therefore, the most critical system state with respect to the supply voltage occurs during the data transmission, when no energy is received by the transponder because of the load modulation. As only digital data are shifted to the ASK transistor during this state, the reset threshold of 940 mV is sufficient for an uninterrupted system operation.

For on-chip timing, the carrier frequency f_c of 13.56 MHz is used as reference, with the modulation frequency f_m of 847.5 kHz and the data frequency f_d of 13.56 kHz being derived as submultiples of f_c by two counters (cf. Fig. 18). f_m is used as the ASK modulation frequency and as clock signal for the digital control logic and the ADC. f_d defines the speed of data transmission. As the data transmission itself is realized by ASK modulation of the received power signal, the frequency spectrum measured at the external RFID reader is composed of a 13.56-kHz data stream modulated with 847.5 kHz at a carrier frequency of 13.56 MHz and convolution products thereof.

While the external reader coil has no strict area limitations, the internal transponder coil is subject to the same limitations as the CMOS chip. For this reason, a planar micro coil of $2 \times 2.5 \text{ mm}^2$ is fabricated. For the windings, gold (Au) is electroplated on a Pyrex wafer into a thick photoresist mask made of AZ 40XT [16]. This step is followed by resist removal and by SU-8 encapsulation. With this process, a track width of $5 \mu\text{m}$ and a height of $20 \mu\text{m}$ are realized. With a distance of $10 \mu\text{m}$ between the windings, 35 windings are realized in the available area (Fig. 7). An inductance of $2 \mu\text{H}$ and capacitance of 1 pF was measured for the fabricated planar micro coil. This results in a self-resonant frequency of 113 MHz. The resistance was measured to be 30Ω . Deviation from the theoretical coil resistance R_{coil} of

$$R_{\text{coil}} = \rho_{\text{Au}} \frac{l_{\text{coil}}}{A_{\text{coil}}} = 25.46 \Omega \quad (2)$$

with $\rho_{\text{Au}} = 22.14 \text{ m}\Omega \times \text{mm}^2/\text{m}$, $l_{\text{coil}} = 115 \text{ mm}$, and $A_{\text{coil}} = 100 \mu\text{m}^2$ could be caused by cross section variations or contact resistances.

The choice of integrating an on-chip coil within the top layer of the employed four-metal-CMOS process was discarded due to the very low quality factor: ignoring the very small effect of wire thickness on inductance L [17], the quality factor $Q = \omega L/R$ of a CMOS integrated coil with the given dimensions is 0.1, while it is 6.7 for the electroplated coil at 13.56 MHz.

CMOS sensor chip and microcoil are electrically connected using thermosonic flip-chip bonding with gold (Au) stud bumps

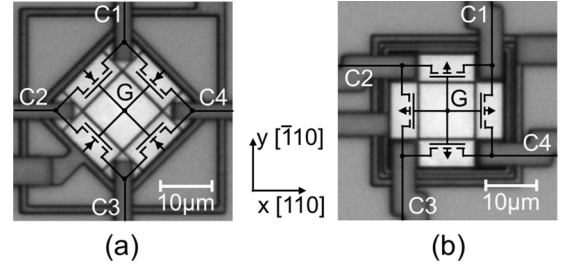


Fig. 8. Micrograph with overlaid transistor diagram of the stress sensors of (a) NMOS and (b) PMOS type.

[18]. Five bumps (four in the corners and one in the center) are placed on the aluminum bond pads of the CMOS chip. While the pad in one corner and in the center are required for the electrical connection, the three additional bumps are added for mechanical symmetry.

B. Sensor Elements

Twenty-four stress sensors for measuring two different stress components are distributed on the surface of the chip [Fig. 1(b) and (c)]. Each of the sensors forms a square of active elements and can be described by four transistors in a Wheatstone bridge arrangement (Fig. 8). The orientation is indicated by the coordinate system (x, y) parallel to the chip edges.

The doping polarity has a strong impact on the piezoresistive coefficients Π_{ij} of MOS inversion layers. For instance, Π_{44} can be ten times higher in P-type than in N-type silicon, while Π_{12} is nearly 50 times higher in the N-type [19]. The in-plane shear stress σ_{xy} therefore is measured by an NMOS type sensor rotated by 45° with respect to (x, y) , while the PMOS type is oriented parallel to the coordinate system and is sensitive to the difference of in-plane normal stresses $(\sigma_{xx} - \sigma_{yy})$. All four transistors within each sensor have a W/L ratio of $5 \mu\text{m}/10 \mu\text{m}$ [20]. These empirically evaluated dimensions represent a good compromise between four main sensor characteristics. Following standard transistor equations, an increased W/L ratio increases the power consumption, while it decreases the noise level. Furthermore, a larger active area reduces the offset, but also reduces the spatial resolution of each sensor and thereby the reliability of the distributed stress mapping itself.

Each sensor can be activated by connecting the common gate of its transistors to V_{DD} for the NMOS and to V_{SS} for the PMOS type. The power consumption of the overall system thus can be reduced, as only the processed sensor is activated temporarily. In [11] and [21], two opposing contacts of the Wheatstone bridge were directly connected to the supply rails, while the remaining two formed the differential sensor output $V_{\perp}$. In the actual version, a single series resistor R_{CM} of $2 \text{ k}\Omega$ is placed between the global V_{SS} rail and the common lower supply of all sensors (Fig. 9), resulting in an increased common-mode voltage V_{CM} for the NMOS sensor, while the operating point of the PMOS is unaltered. This shift in V_{CM} matches the two sensor common mode signals to the input range of the DDA. Implementing R_{CM} only for the NMOS sensors would have required massive design and layout changes, as biasing and

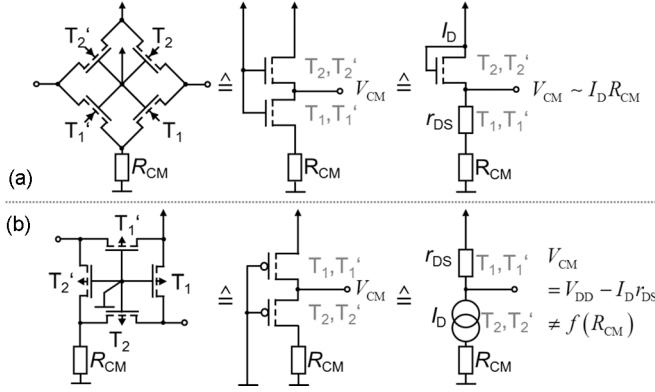


Fig. 9. Sensor schematic (activated) showing the influence of the common mode resistor R_{CM} on (a) NMOS and (b) PMOS sensors.

routing were laid out with a common supply grid for all 24 sensors distributed over the entire chip.

The equivalent model shown in Fig. 9(a) describes the NMOS sensor as a series connection of R_{CM} and of the parallel-connected NMOS transistors with the center tap at V_{CM} . Assuming an operation in the linear region for the lower transistor part ($T_1, T_{1'}$) and saturation for the upper ($T_2, T_{2'}$), the current through the lower and upper part can be expressed, respectively, as

$$I_{T_1, T_{1'}} = \beta \left(V_{DD} - V_{R_{CM}} - V_{th} - \frac{V_{CM} - V_{R_{CM}}}{2} \right) (V_{CM} - V_{R_{CM}})$$

$$I_{T_2, T_{2'}} = \frac{\beta}{2} (V_{DD} - V_{CM} - V_{th})^2 \quad (3)$$

where β is the transistor gain factor $\mu_n \epsilon_{ox} / t_{ox}$ times the ratio of transistor dimensions W/L . As these two currents have to be equal, the common-mode voltage V_{CM} at the center tap of the sensor can be derived as

$$V_{CM} = (V_{DD} - V_{th}) \left(1 - \frac{1}{\sqrt{2}} \right) + \frac{V_{R_{CM}}}{\sqrt{2}} \quad (4)$$

by neglecting any second-order effects, in particular the body effect. V_{CM} of the NMOS sensor is thus dependent on the voltage drop at the common-mode resistor $V_{R_{CM}}$, that is calculated as

$$V_{R_{CM}} = (V_{DD} - V_{th}) + \frac{1}{\beta R_{CM}} - \sqrt{\frac{2V_{DD}}{\beta R_{CM}} + \left(V_{th} - \frac{1}{\beta R_{CM}} \right)^2} \quad (5)$$

The equivalent model for the PMOS sensor [Fig. 9(b)] reveals an operation independent of R_{CM} , as the architecture can be described as a PMOS current source ($T_2, T_{2'}$) with a PMOS resistor ($T_1, T_{1'}$) at its source—forming a Widlar current source—and a resistive load (R_{CM}) at its drain. As long as the resistive load is small compared to the internal resistance of the current source, a constant current will be generated, leading to a constant voltage drop at $T_1, T_{1'}$ almost independent of R_{CM} .

The sensing effect of both types of sensors is related to the piezoresistive coefficients. The differential voltage $V_{\perp}$ between the two central contacts of the Wheatstone bridge is scaled with combinations of the piezoresistive coefficients termed Π_{11}, Π_{12}

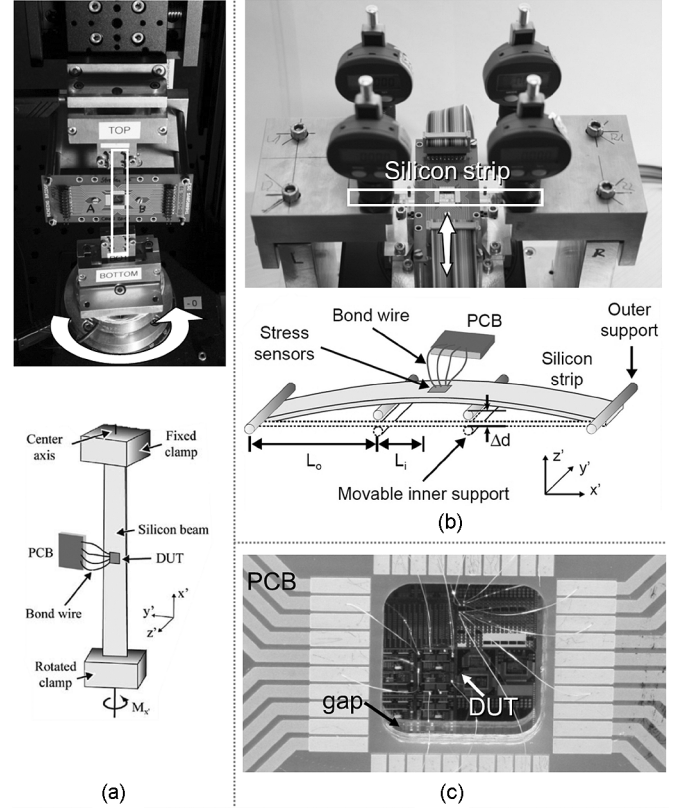


Fig. 10. Photograph of (a) the torsion bridge and (b) the four-point bending. White boxes mark the position of the wafer strip and white arrows show the direction of movement of the bridge. (c) Bond wire connection from the DUT to the PCB with negligible mechanical contact. The schematics of bending and torsion bridge are taken from [22] and [23], respectively.

and Π_{44} of inversion layers, expressing a second-order dependency on the angle φ between the x-axis and the current direction [21] as

$$V_{\perp} \propto (\Pi_{11} - \Pi_{12}) \sigma_{xy} \cos(2\varphi) - \frac{\Pi_{44}}{2} (\sigma_{xx} - \sigma_{yy}) \sin(2\varphi). \quad (6)$$

Since the sensors are laid out in a fourfold rotational symmetry, the current and sensing directions are rotated in four steps by 90° by an analog MUX. According to [21], this discrete current switching (DCS) operation enables the elimination of several non-mechanical signal contributions, e.g., temperature gradients and magnetic fields. By using the DCS operation and naming the voltages resulting from the four measurements for each sensor direction $V_{\perp, i}$, the sensor signal proportional to the in-plane shear stress σ_{xy} is calculated as [21]

$$V_{\perp, \sigma_{xy}} = \frac{V_{\perp, 1} - V_{\perp, 2} + V_{\perp, 3} - V_{\perp, 4}}{4}. \quad (7)$$

Each sensor is located at a point of maximum stress insertion of its respective sensitivity type (σ_{xx-yy} or σ_{xy}). These optimal sensor positions were verified by FE simulations with a realistic smart bracket model [8]. Furthermore, low-stress areas were identified for the circuitry, especially for the analog elements of the sensor readout.

For the stress sensor characterization, four-point bending and torsion bridges [22]–[24] were used to apply well-defined in-plane normal σ_{xx} and shear stress σ_{xy} , respectively (Fig. 10).

In order to extract the sensitivity, a wafer strip of approximately $9 \times 110 \text{ mm}^2$ containing the device under test (DUT) in the center is mechanically loaded, while the electrical response of the DUT is measured. In Fig. 10(c), an enlargement of the DUT and its connection to the PCB are shown. During the measurements, the strip is released from the PCB and therefore only the thin bond wires introduce a negligible mechanical error stress to the DUT.

Both implemented sensor types are characterized with the standard process supply voltage of 3.3 V. The resistor R_{CM} in the negative supply rail shows only an influence on the NMOS sensors, as expected from theory. The measurements without R_{CM} are noted in brackets. From a set of five samples, absolute sensitivities of $-836 \pm 2.44 \mu\text{V}/\text{MPa}$ ($-978 \mu\text{V}/\text{MPa}$) for NMOS devices to σ_{xy} and $921 \pm 4.74 \mu\text{V}/\text{MPa}$ ($921 \mu\text{V}/\text{MPa}$) for PMOS devices to σ_{xx} are determined [Fig. 11(a) and (b)]. The average current consumption is $153 \mu\text{A}$ ($209 \mu\text{A}$) for NMOS and $47 \mu\text{A}$ ($47 \mu\text{A}$) for PMOS sensors. Furthermore, the sensitivity was monitored to be linearly dependent on small variations of the supply voltage: The sensitivity changes by $-173 \mu\text{V}/\text{MPa}/\text{V}$ ($-237 \mu\text{V}/\text{MPa}/\text{V}$) for NMOS and $347 \mu\text{V}/\text{MPa}/\text{V}$ ($347 \mu\text{V}/\text{MPa}/\text{V}$) for PMOS sensors. The cross sensitivities for the undesired stress component result in insignificant errors. Due to the tedious measurements with the silicon strips, only a set of three samples was used for the cross sensitivity characterization, and values of $11.4 \pm 13.6 \mu\text{V}/\text{MPa}$ for NMOS devices to σ_{xx} and $15.7 \pm 14.0 \mu\text{V}/\text{MPa}$ for PMOS devices to σ_{xy} were extracted. These cross sensitivities constitute approximately 1.5% of the main sensitivities and thus are considered negligible in comparison to the aimed stress resolution of 10% of the respective nominal value.

Two sensor parameters with great importance for the configuration of the sensor readout electronics are the common mode level V_{CM} and the intrinsic offset $V_{\perp, \text{off}}$. V_{CM} is defined as the average potential of the two measurement contacts with respect to V_{SS} . Average values for V_{CM} are $929 \pm 2.2 \text{ mV}$ (721 mV) for the NMOS and $2.608 \pm 1.8 \text{ mV}$ (2.608 mV) for the PMOS sensors. $V_{\perp, \text{off}}$ is the differential voltage of the measurement contacts without externally applied mechanical stress. It is caused by geometrical imperfections and residual stress introduced during fabrication. As expected from the theoretical sensor behavior described in (6), the signals $V_{\perp, i}$, from biasing directions 0 and 2 are almost identical. The same is true for directions 1 and 3 with inverted sign compared to directions 0 and 2. The shear stress related offset voltage $V_{\perp, \text{off}, \sigma_{xy}}$ after DCS compensation according to (7) is $-0.62 \pm 1.57 \text{ mV}$ for the NMOS sensors, the normal stress related $V_{\perp, \text{off}, \sigma_{xx}}$ is $8.62 \pm 3.02 \text{ mV}$ for the PMOS sensors. While the shear stress component is small compared with the standard deviation, the relatively high normal stress component originates most likely from residual stress from handling and fabrication, e.g., single-sided deposition of passivation layers. A second cause is the pressure applied by the probe-card needles. The average force exerted by one probe needle during the characterization was measured to be 35 mN . An influence of additional force was only observed at the PMOS sensors where this force causes an additional offset voltage of $\pm 1.9 \text{ mV}$, which is equal to a normal stress of $\approx 2.2 \text{ MPa}$.

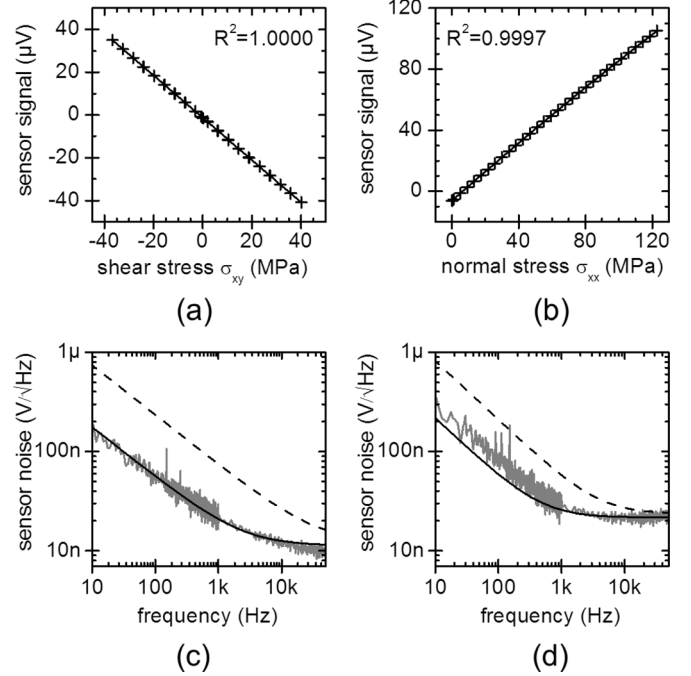


Fig. 11. (a), (b) Measured sensor response to well-defined stresses and (c), (d) noise characteristic for the NMOS and the PMOS sensors. The solid gray and black lines represent the measurements and BSIM simulations, respectively. The dashed lines represent the noise prediction for the standard supply voltage of 3.3 V.

The noise performance of the sensors is not characterized at the intended operating point with $V_{DD} - V_{SS} = V_G = 3.3 \text{ V}$ as the intrinsic sensor offset in combination with the gain of the available discrete OP37 low-noise preamplifier of 9000 causes signal clipping. $V_{DD} - V_{SS} = 1.0 \text{ V}$ for the NMOS and 0.5 V for the PMOS sensor was chosen instead to reduce the sensitivity, while $V_G = 3.3 \text{ V}$ for both sensor types. Fig. 11(c) and (d) presents the measured results and the prediction of BSIM simulations for the four-transistor sensor models from Fig. 8. We conclude from the high compliance of simulation and measurement that the noise prediction for the standard supply voltage of 3.3 V will be sufficient for a simulation-based system design. The simulation-based prediction for the integrated output noise of the NMOS and PMOS sensor including the common-mode resistor is $8.1 \mu\text{V}_{\text{rms}}$ and $8.4 \mu\text{V}_{\text{rms}}$, respectively, integrated from 1 Hz to 36 kHz, which is the frequency band analyzed by the sensor readout. Taking the respective sensitivities into account, these values correspond to noise levels of 9.7 kPa for NMOS and 9.1 kPa for PMOS sensors.

C. Sensor Readout

The sensor readout is composed of a differential difference amplifier (DDA) with variable gain, followed by a SAR ADC and a gain controller (GC) (Fig. 12). Based on the digitized sensor value, the GC determines the maximum ideal gain value with respect to the linear DDA output range on the basis of a binary tree search.

The symmetry of the sensors results in a high linearity, while their output impedance is limited by the sensor components which are in the linear region (T_1, T_1' in Fig. 9). Thus cross

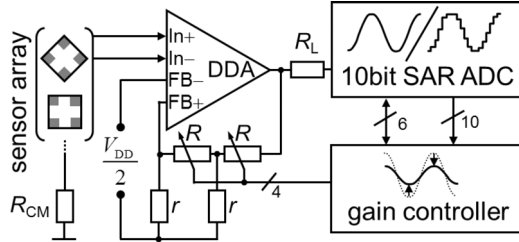


Fig. 12. Topology of the sensor readout including DDA, ADC, and gain controller.

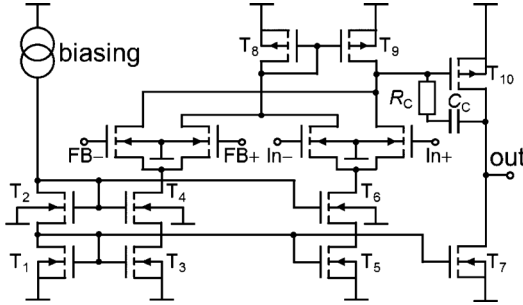
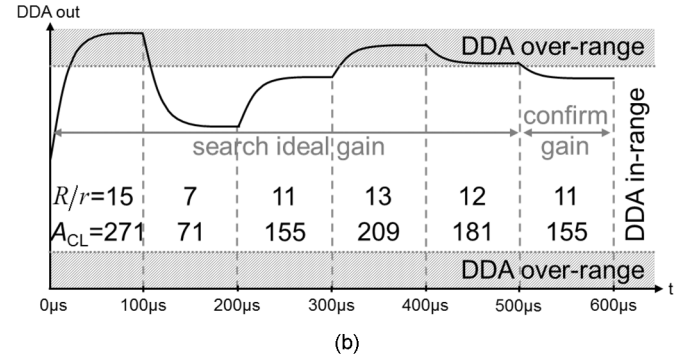
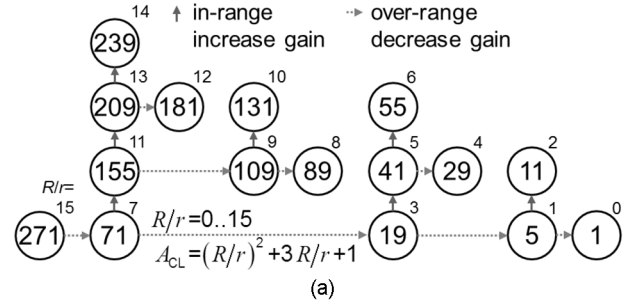


Fig. 13. Transistor diagram of the DDA.

currents out of the sensor should be avoided to prevent influence on the operating point. The architecture of the DDA offers one differential input pair for the sensors and a separate pair for the feedback. Thus, a high-ohmic interface for the sensor connection is provided and impedance variations in the feedback path during gain variation are not influencing the load impedance of the sensor during readout. Compared to conventional 3-Opamp instrumentation amplifiers, the DDA achieves an improved offset performance due to better matching potential of the input stage, when a fourfold common centroid arrangement is used. Operating the sensors in the DCS mode according to (7) furthermore permits the use of an unchopped amplifier; rotating the current in the sensor results in compensation of several non-mechanical signal contributions, e.g., amplifier offset and temperature gradients [21].

Fig. 13 shows the transistor diagram of the DDA. All current source transistors ($T_3 - T_7$) are sized with $W/L = 32 \mu\text{m}/4 \mu\text{m}$ and deliver $25 \mu\text{A}$. When compared with the input current sources, the output source omits the cascode transistors for the benefit of a high-output swing. Relatively large transistors of $W/L = 100 \mu\text{m}/5 \mu\text{m}$ are chosen for the differential input pair T_{in} and feedback pair T_{FB} for noise considerations: independent of the gain, the total input-referred noise of the DDA has to be in the range of the noise coming from the sensing devices. The remaining transistors $T_8 - T_{10}$ are sized with $W/L = 80 \mu\text{m}/5 \mu\text{m}$. The output stage is Miller compensated for stability reasons with $R_C = 23 \text{ k}\Omega$ and $C_C = 4.5 \text{ pF}$.

The closed-loop gain can be selected digitally in 16 steps ranging from 1 to 271. The aimed stress resolution of 25 kPa measured by an NMOS sensor with $836 \mu\text{V}/\text{MPa}$ thus can be amplified up to 5.66 mV , which is nearly 2 LSB of the used 10-bit ADC. In addition, the buffer configuration offers the sensing of signals even above 120 MPa , which was empirically


 Fig. 14. (a) Possible gain values arranged in a binary search tree. Small numbers correspond to the resistor ratio R/r . (b) Example for the gain evaluation algorithm.

evaluated as the maximum uncritical stress for the CMOS wafer stripes.

A resistive T-network is used for feedback variation, as it achieves higher gains with smaller resistance ratios compared to conventional resistive dividers (Fig. 12) [25]. The closed-loop gain A_{CL} is determined by the resistance ratio R/r with values between 0 and 15 and $r = 109 \text{ k}\Omega$ as

$$A_{CL} = \left(\frac{R}{r}\right)^2 + 3\frac{R}{r} + 1. \quad (8)$$

Assuming the gain to be changed from 1 to 271, the impedance of a conventional resistive divider would be changed by a factor of 271. For the T-network it only changes by a factor of 32 from $r_{out}|_{R=0r} = r \parallel r = (r)/(2)$ to $r_{out}|_{R=15r} = R + r \parallel (R + r) \approx 16r$, leading to decreased load impedance variations at the DDA's output stage. Furthermore, the T-network allows for an improved element matching and reduced area consumption.

Fig. 14(a) lists all 16 gain values with their respective R/r ratios arranged in a binary search tree, as it is used during the gain evaluation process. The possibility of gain variation and the automatic adjustment based on the sensor values provides an increased dynamic range. Large sensor signals could exceed the DDA's linear output range after amplification and therefore need to be reduced in gain as described in the following.

After analog amplification the sensor signal is digitized by a standard 10-bit SAR ADC ([12], aadcc02). With an output common-mode voltage of 1.65 V representing zero stress, a digital value below $79 \text{ LSB} \hat{=} 255 \text{ mV}$ or above $(1024 - 79) \text{ LSB} \hat{=} 3.045 \text{ V}$ presents an amplified signal out of the DDA's linear range ("over-range"), leading to a reduction of the gain. If the amplified signal is between these borders ("in-range") the gain

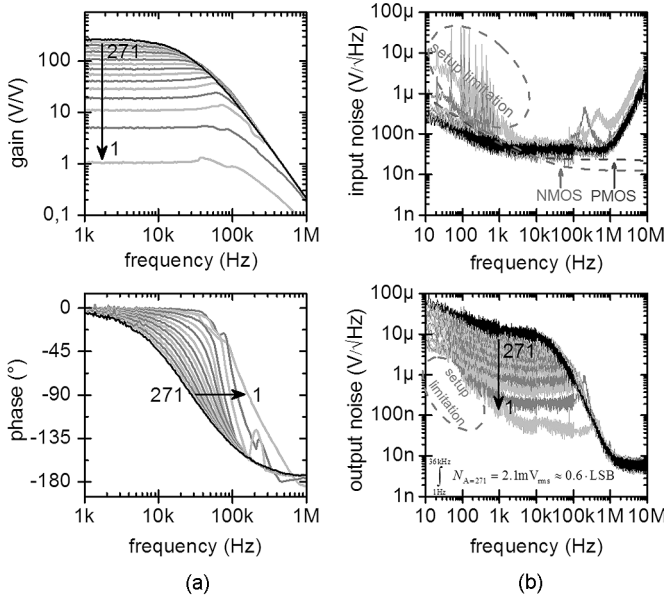


Fig. 15. (a) Measured Bode plot for the DDA for all 16 gain values and (b) the input and output referred noise spectral density.

is increased. The borders of the range have been evaluated by simulations. To determine the ideal amplification, the GC iteratively interprets the digitized amplified sensor signal, adjusts the gain and re-initiates the readout for up to six times. With the re-initialization occurring every 100 μs the sensed dc stress signal is frequency shifted to 10 kHz.

Fig. 14(b) gives an example of one sensor readout based on the binary search tree from Fig. 14(a). The first readout is always performed with the maximum gain of 271. In the case of “over-range,” the search algorithm takes 500 μs to evaluate the maximum gain within the linear output range. In this example, the ideal gain of 155 is found in the third readout attempt, as the fourth as well as the fifth result in “over-range” signals. The sixth readout confirms this gain and averages both results to suppress disturbances. In case of “in-range” during the first readout with a gain of 271 the averaging is executed directly without gain evaluation and the readout is completed after 200 μs . An error flag is set if the second readout with “ideal” amplification after gain examination is in the ‘over-range’. After the gain is evaluated successfully, the averaged 10-bit sensor value together with a 5-bit sensor number, 2-bit bias direction, the error flag, and 4-bit determined gain is stored in a 22-bit serial shift register for data transmission.

The DDA was designed for a GBW of 2 MHz and an open-loop dc gain of 92 dB. The load resistor $R_L = 436 \text{ k}\Omega$ together with the ADC’s input capacitance of 10 pF (Fig. 12) introduce an additional pole at 36 kHz which suppresses high-frequency artifacts above the shifted stress signal at 10 kHz. Fig. 15(a) shows the measured closed-loop Bode plot for all 16 gain values of the DDA with activated ADC. All gain and phase values are in very good agreement with simulation results. The input-referred offset voltage of three separately characterized DDAs varies from 38 to 270 μV . While the upper input common mode limit of all devices was beyond V_{DD} , the lower ranged from 690 to 800 mV. This variation in combination with a native sensor

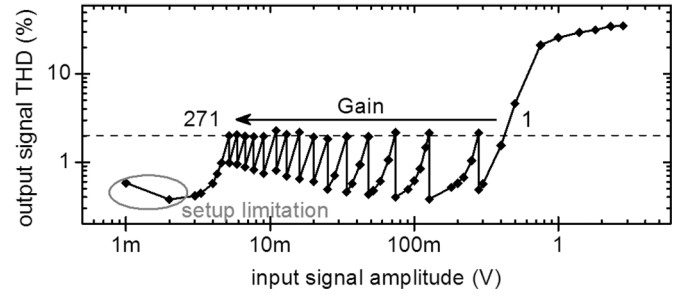


Fig. 16. Measured linearity of the DDA showing the effect of the GC for all 16 gain values.

common mode V_{CM} of 721 mV for NMOS sensors required the previously introduced shifting resistor R_{CM} to guarantee a successful readout.

The measured input-referred noise spectral density presented in Fig. 15(b) is nearly constant for all gain values and matches the simulation results. The integrated input-referred noise in a frequency band from 1 Hz to 36 kHz (which is the input pole of the ADC) calculates to 10.8 μV_{rms} and thus slightly dominates the sensor noise of less than 8.4 μV_{rms} [dotted lines in Fig. 15(b)]. The integrated output noise of the DDA is approximately proportional to the selected amplification and the maximum gain of 271 results in an average output noise voltage of 2.1 mV_{rms} .

The measured total harmonic distortion (THD%) of the DDA is shown in Fig. 16. The gain is stepwise decreased with increasing input signal, leading to a maximum THD% of approximately 2% until the input signal exceeds 400 mV. The gain controller thus generates a compromise between linearity and resolution for appropriate input signals.

Extensive measurements show that the timing and gain evaluation of the complete readout system (Fig. 12) are functional for the full range of input signals and the 22-bit data stream is generated successfully. The sensor readout consisting of DDA, ADC, and GC consumes 290 μA at 3.3 V.

For the overall noise characterization, the standard deviation of the digital data stream is measured at the output of the ADC. Without activated sensors, the deviation is caused by the DDA and the ADC. For a dc supplied system the standard deviation was measured as 0.9 LSB, which increased to 1.9 LSB with wireless power supply. For the 10-bit ADC supplied with 3.3 V, this results in an input-referred noise of 2.9 mV for dc and 6.1 mV for wireless supply. As the standard deviation was measured to be independent of the selected gain of the DDA, the authors conclude that, even for maximum gain, the overall output noise is dominated by the ADC and not by the DDA with its previously presented maximum output noise of 2.1 mV_{rms} . The standard cell ADC therefore represents the system’s bottleneck in terms of noise performance and should be replaced in future versions.

IV. SYSTEM CHARACTERIZATION

Fig. 17(a) shows the complete stress mapping chip with full functionality in a chip area of $2 \times 2.5 \text{ mm}^2$. Fig. 17(b) presents the assembly of chip and microcoil, which is 0.73 mm high. For evaluation of the wireless operation, the stress mapping

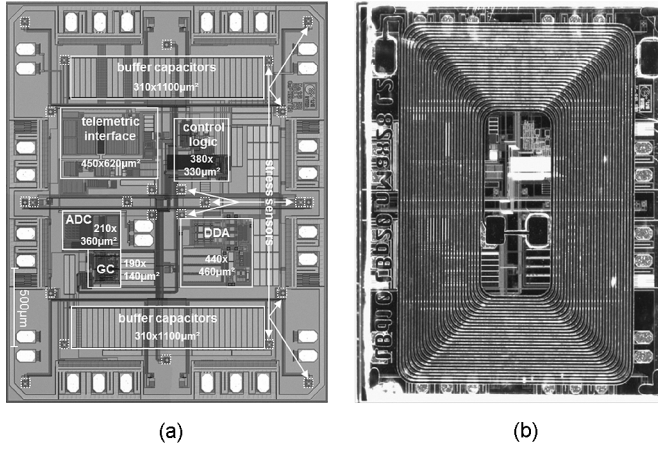


Fig. 17. (a) Micrograph of the stress mapping chip. The 24 sensors are placed at the most stress sensitive positions, the signal processing at relatively low stress areas. (b) Assembly of chip and microcoil.

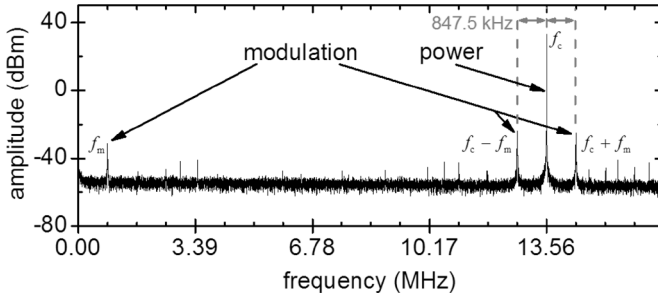


Fig. 18. Frequency spectrum during wireless operation, measured at the excitation coil.

system is excited by an external 1.8- μ H SMD coil from Murata's LQW series. All measurements in this chapter are based on a supply voltage of 3.19 V, defined by the fine limiter presented in Section III-A.

The power required to wake up the stress sensing system is 45 mW (measured with a shunt in series to the micro coil) transmitted by the external unit at 13.56 MHz. Thus, the rectifier based on the negative voltage converter [14] improved the conventional full bridge approach by reducing the required power by a factor of 3, compared to the 138 mW required in [11]. After start-up and as long as a minimum of 17 mW are available, the system remains operational and alternates between the data acquisition, data sending, and buffer recharging states. A received power of 250 mW was measured with the aforementioned setup at direct contact between the primary and the secondary coil. Up to a distance of 1 mm the system was fully operational, even though the supply voltage dropped to approximately 2.3 V during the data sending phase. The small communication distance is caused by the actually untrimmed micro coil. Its self-resonant frequency of 113 MHz can be reduced by a capacitor array in future designs, offering more practical communication distances for orthodontic treatment of several millimeters. At the external reader coil power and data are clearly separated into two channels by the modulation frequency f_m (Fig. 18). As $f_m = 847.5$ kHz is generated on-chip as submultiple of the carrier frequency, it illustrates the success of wireless operation.

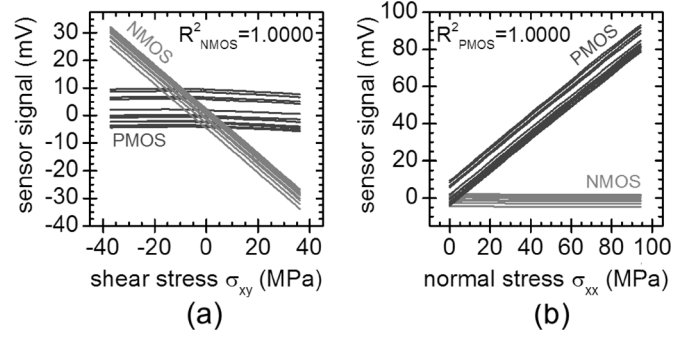


Fig. 19. Measured system response to (a) normal stress and (b) shear stress.

Excluding the wireless interface, the rest of the system i.e., the sensor biasing, the MUX, the readout including ADC and gain evaluation, as well as the digital control of the system, consumes an average of 1.75 mW at 3.19 V. Each sensor thereby is activated once per processing cycle and read out for each of its four current directions (see also Fig. 3). A maximum duration of 400 ms for the readout and data transmission of all 24 sensors was measured. This is in perfect agreement with the expected cycle time, when each sensor evaluation consists of 6 readout attempts.

The complete system was exposed to defined mechanical stress by the setup described in Section III-B. The measurement results are presented in Fig. 19(a) for the torsion and Fig. 19(b) for the bending bridge. All sensor elements are responding linearly ($R^2_{\text{NMOS}} > 0.99998$, $R^2_{\text{PMOS}} > 0.99996$) and with identical sensitivity to their respective stress component. Sensitivities of $-783 \mu\text{V/MPa}$ for the NMOS and $904 \mu\text{V/MPa}$ for the PMOS sensors were measured. Normalizing to the supply voltage used in the measurements, the sensitivities of the PMOS sensors correspond to those of the sensor characterization in Section III-B. However, the supply-normalized NMOS sensitivity is 4.1% below the expected value. This difference is most probably caused by an imprecise adjustment of the wafer strip on the torsion bridge. Not only the NMOS sensitivity is decreased but also the PMOS sensors are exposed to small differences of normal stress. For positive shear stresses, the PMOS respond with a sensitivity of 3.5% of the nominal value expected for the NMOS sensors (Fig. 19(a), $\sigma_{xy} > 0$). The externally applied and measured shear stress thus seems to be translated into a combination of normal and shear stress detectable at the DUT. The cross sensitivities of $10 \mu\text{V/MPa}$ for the NMOS to σ_{xx} and of $21 \mu\text{V/MPa}$ for the PMOS device to σ_{xy} (for negative shear stress values only) are comparable to the pure sensor characterization.

Based on the ADC quantization error of 1.9 LSB for wireless supply and the 8-fold measurement of each sensor (once for all four directions, twice during each gain evaluation), a maximum input-referred resolution of $8 \mu\text{V}$ is expected for maximum gain. To verify the resolution by measurements, the standard deviation of each sensor to its respective linear regression is used. The regression thereby not only eliminates the offset signal of each sensor but also of the readout electronics. For both sensor designs, an uncertainty of $10 \mu\text{V}$ is revealed close to zero stress, i.e., the operating region with maximum gain. This corresponds

TABLE I
SPECIFICATIONS OF THE WIRELESS STRESS MAPPING SYSTEM

technology	0.35 μm CMOS & Au electroplating on Pyrex	
system size	$2 \times 2.5 \times 0.73 \text{ mm}^3$	
supply voltage	3.3 V (3.19 V)	
power consumption	1.75 mW (w/o RF interface)	
communication approach	inductive coupling with ASK	
communication frequency	$f_c = 13.56 \text{ MHz}$ $f_m = 847.5 \text{ kHz}$	(power link) (ASK modulation)
sensor principle	24 piezoresistive FET 10 $\times$ NMOS, 14 $\times$ PMOS	
sensitivity	$-783 \mu\text{V}/\text{MPa}$ to σ_{xy} $904 \mu\text{V}/\text{MPa}$ to σ_{xx}	(NMOS) (PMOS)
cross-sensitivity	$11.4 \mu\text{V}/\text{MPa}$ to σ_{xx} $15.7 \mu\text{V}/\text{MPa}$ to σ_{xy}	(NMOS) (PMOS)
resolution	10 bit $\equiv 10 \mu\text{V} - 2.71 \text{ mV}$ $\equiv 13 \text{ kPa} - 3.46 \text{ MPa}$ $\equiv 11 \text{ kPa} - 3.00 \text{ MPa}$	($A_{CL} = 271 - 1$) (NMOS) (PMOS)
linearity	THD% < 2%	
sample time	3.81 ms – 4.21 ms per sensor direction 366 ms – 404 ms per system cycle	

to a maximum resolution of 13 kPa for the NMOS and 11 kPa for the PMOS devices in a calibrated system, which satisfies the requirements of 25 kPa for the smart bracket application.

V. CONCLUSION

A wireless stress mapping system with a volume of $2 \times 2.5 \times 0.73 \text{ mm}^3$ is presented. Such chip and coil dimensions enable encapsulation into the basis of an orthodontic bracket. 24 stress sensors sensitive to either in-plane shear stress or the difference of in-plane normal stresses are implemented at strategically optimal positions on the chip surface using a 0.35- μm CMOS process. The system achieves a maximum stress resolution of 11 kPa with a power consumption of 1.75 mW for the stress evaluation. Further specifications are listed in Table I. With its performance values, the system paves the way towards the implementation of smart brackets in orthodontic therapy, providing the clinician with direct sensory feedback on the forces and moments applied to individual teeth. Such feedback may increase treatment effectiveness and reduce negative therapeutic side effects such as irreversible root resorption due to overloading of teeth.

REFERENCES

- [1] J. Suhling and R. Jaeger, "Silicon piezoresistive stress sensors and their application in electronic packaging," *Sensors J.*, vol. 1, pp. 14–30, 2001.
- [2] J. Roberts, S. Hussain, M. K. Rahim, M. Motalab, J. C. Suhling, R. C. Jaeger, P. Lall, and R. Zhang, "Characterization of microprocessor chip stress distributions during component packaging and thermal cycling," in *Proc. 60th Electron. Compon. Technol. Conf.*, Jun. 2010, pp. 1281–1295.
- [3] M. Herrmann, P. Gieschke, P. Ruther, and O. Paul, "Cmos-integrated three-axis force sensor for coordinate measurement applications," in *Proc. IEEE Sensors*, Nov. 2010, pp. 2648–2652.
- [4] C. Peters, B. Pommerehne, R. Gutwald, R. Schmelzeisen, and Y. Manoli, "A wireless in-vivo stress measurement system in osteosynthesis," in *Proc. Int. Conf. Biomed. Eng.*, Dec. 2005, pp. 4B1–01.
- [5] B. G. Lapatki, J. Bartholomeyczik, P. Ruther, I. E. Jonas, and O. Paul, "Smart bracket for multi-dimensional force and moment measurement," *J. Dental Res.*, vol. 86, pp. 73–78, 2007.
- [6] M. C. Meikle, "The tissue, cellular, and molecular regulation of orthodontic tooth movement: 100 years after Carl Sandstedt," *Eur. J. Orthodontics*, vol. 28, no. 3, pp. 221–240, 2006 [Online]. Available: <http://ejo.oxfordjournals.org/content/28/3/221.abstract>
- [7] E. Chan and M. A. Darendeliler, "Physical properties of root cementum: Part 5. volumetric analysis of root resorption craters after application of light and heavy orthodontic forces," *Amer. J. Orthodontics Dentofacial Orthopedics*, vol. 127, no. 2, pp. 186–195, 2005.
- [8] S. Rues, B. Panchaphongsaphak, P. Gieschke, O. Paul, and B. Lapatki, "An analysis of the measurement principle of smart brackets for 3d force and moment monitoring in orthodontics," *J. Biomech.*, vol. 44, no. 10, pp. 1892–1900, 2011 [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0021929011003496>
- [9] P. Gieschke, Y. Nurcahyo, M. Herrmann, M. Kuhl, P. Ruther, and O. Paul, "Cmos integrated stress mapping chips with 32 n-type or p-type piezoresistive field effect transistors," in *Proc. IEEE 22nd Int. Conf. Micro Electro Mech. Syst.*, Jan. 2009, pp. 769–772.
- [10] P. Gieschke and O. Paul, "Cmos-integrated sensor chip for in-plane and out-of-plane shear stress," in *Proc. Eurosensor XXIV Conf.*, Sept. 2010, vol. 5, pp. 1364–1367 [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S187770581000915X>
- [11] M. Kuhl, P. Gieschke, D. Rossbach, S. A. Hilzensauer, P. Ruther, O. Paul, and Y. Manoli, "A telemetric stress-mapping cmos chip with 24 fet-based stress sensors for smart orthodontic brackets," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2011, pp. 108–110.
- [12] "XH035- 0.35 Micron Modular Analog Mixed Signal Technology With RF Capability and HV Extensions," X-FAB Semiconductor Foundries, Erfurt, Germany, Mar. 2011 [Online]. Available: http://www.xfab.com/fileadmin/X-FAB/Download_Center/Technology/CMOS/XH035_HVCMOS_RFCMOS_Data_sheet.pdf
- [13] K. Finkenzeller, *RFID-Handbuch*, 3rd ed. Munich, Germany: Carl Hanser Verlag, Sept. 2002.
- [14] C. Peters, D. Spreemann, M. Ortmanns, and Y. Manoli, "A cmos integrated voltage and power efficient ac/dc converter for energy harvesting applications," *J. Micromech. Microeng.*, vol. 18, no. 10, pp. 104005–, 2008 [Online]. Available: <http://stacks.iop.org/0960-1317/18/i=10/a=104005>
- [15] M. Ghovanloo and K. Najafi, "Fully integrated wideband high-current rectifiers for inductively powered devices," *IEEE J. Solid-State Circuits*, vol. 39, no. 11, pp. 1976–1984, Nov. 2004.
- [16] "AZ 40XT -Thick Positive Photoresist," MicroChemicals GmbH, 2012 [Online]. Available: http://www.microchemicals.com/photore-sist/thick/_positive/_photoresist.html
- [17] S. Mohan, M. del Mar Hershenson, S. Boyd, and T. Lee, "Simple accurate expressions for planar spiral inductances," *IEEE J. Solid-State Circuits*, vol. 34, no. 10, pp. 1419–1424, Oct. 1999.
- [18] J. H. Lau, J. H. Lau, Ed., *Flip Chip Technologies*. New York, NY, USA: McGraw-Hill, 1995.
- [19] R. Jaeger, J. Suhling, R. Ramani, A. Bradley, and J. Xu, "CMOS stress sensors on [100] silicon," *IEEE J. Solid-State Circuits*, vol. 35, no. 1, pp. 85–95, Jan. 2000.
- [20] P. Gieschke, B. Sbiński, and O. Paul, "Cmos-based piezo-fet stress sensors in wheatstone bridge configuration," in *Proc. IEEE Sensors*, Oct. 2011, pp. 93–96.
- [21] M. Kuhl, P. Gieschke, O. Paul, and Y. Manoli, "A differential difference amplifier with automatic gain selection as readout interface for cmos stress sensors in orthodontic brackets," in *Proc. 16th Int. Solid-State Sensors, Actuators and Microsyst. Conf.*, Jun. 2011, pp. 819–822.
- [22] P. Gieschke, Y. Nurcahyo, M. Herrmann, M. Kuhl, P. Ruther, and O. Paul, "Integrated stress mapping chip with 32 piezoresistive field effect transistors," in *Proc. Eurosensor XXII Conf.*, Sep. 2008, pp. 292–295.
- [23] M. Herrmann, P. Gieschke, P. Ruther, and O. Paul, "Torsional bridge setup for the characterization of integrated circuits and microsensors under mechanical shear stress," *Rev. Scientific Instrum.*, vol. 82, pp. 1–9, 2011.
- [24] M. Herrmann, P. Gieschke, Z. Liu, J. Korvink, P. Ruther, and O. Paul, "Design and characterization of in-plane silicon stress sensors with isotropic sensitivity," in *Proc. IEEE Sensors Conf.*, Oct. 2008, pp. 1528–1531.
- [25] R. Wu, F. Lidgley, and K. Hayatleh, "Frequency performance compensation of operational amplifiers with the 't' feedback network," *Analog Integr. Circuits Signal Process.*, vol. 41, pp. 79–83, 2004 [Online]. Available: <http://dx.doi.org/10.1023/B:ALOG.0000038285.99051.56>



Matthias Kuhl (S'07) was born in Wuppertal, Germany, in 1981. He received the B.Sc. and M.Sc. degrees in electrical engineering from the University of Wuppertal, Wuppertal, Germany, in 2004 and 2006, respectively. He is working toward the Ph.D. degree at the University of Freiburg, Freiburg, Germany, under the guidance of Prof. Manoli.

Since October 2006, he is with the Fritz Huettinger Chair of Microelectronics at the Department of Microsystems Engineering (IMTEK), University of Freiburg, Freiburg, Germany. His main

field of research is the design of modular microsystems and the integration of "More-than-Moore" components, i.e., electro-chemical fuel cells and electro-mechanical stress sensors. Since January 2013, he heads the group of medical microelectronic devices and implantable circuits at the Fritz Huettinger Chair of Microelectronics.

Mr. Kuhl is student member of VDE.



Pascal Gieschke received the Dipl.-Ing. degree in microsystems engineering from the University of Freiburg, Freiburg, Germany, in 2005.

In 2003–2004, he was a Visiting Student with Portland State University, Portland, OR, USA for nine months. For this stay, he was awarded a tuition waiver and a general scholarship by the German Academic Exchange Service (DAAD). Between 2005 and 2012, he was a Ph.D. student at the Microsystem Materials Laboratory, Department of Microsystems Engineering (IMTEK), University

of Freiburg, Germany. His work focused on the design, characterization and packaging of miniaturized, CMOS-based mechanical stress sensors and sensor systems, in particular for biomedical applications. In 2012, he joined Robert Bosch GmbH, Reutlingen, Germany as an R&D engineer.

Mr. Gieschke is a member of VDI and VDE.



Daniel Rossbach was born in Karlsruhe, Germany, in 1978. He received the Dipl.-Ing. degree in microsystems engineering from the University of Freiburg, Freiburg, Germany, in 2008. His diploma thesis, performed at the Fraunhofer Institute for Applied Solid State Physics, dealt with the design of UHF low-noise amplifiers for the use in magnetic resonance imaging devices.

Since 2009, he has been with the Institute of Micromachining and Information Technology (HSG-IMIT), Villingen-Schwenningen, Germany,

and develops CMOS integrated analog RF circuits for telemetry power and data transfer in biomedical applications at the Fritz Huettinger Chair of Microelectronics under the direction of Prof. Dr.-Ing. Manoli.



Sascha A. Hilzensauer was born in Homburg, Germany, in 1982. He received the Diploma degree from the University of Kaiserslautern, Kaiserslautern, Germany, in 2007, and the M.Sc. degree from the Universities of Freiburg, Freiburg, Germany, in 2010, both in microsystems engineering.

He joined m2k-laser GmbH in 2010, where he is responsible for process development of GaAs- and GaSb-based high-power and high-brightness diode lasers. His main focus lies on plasma etching, plasma deposition, and lithography processes.



Thanapon Panchaphongsaphak was born in Bangkok, Thailand, in 1977. He received the B.Eng. degree (first class honors) in mechanical engineering from Kasetsart University, Bangkok, in 1997, the M.Eng. degree in manufacturing systems engineering from the Asian Institute of Technology, Pathumthani, Thailand, in 2000, the M.S. degree in computational mechanics from Technical University of Munich, Munich, Germany, in 2003, and the Ph.D. degree of mechanical and process engineering from ETH Zurich, Zurich, Switzerland, in 2008.

After postdoctoral works between 2008–2011 at University of Freiburg and University of Ulm, Germany, he returned to Thailand and has been working as a Researcher with the Medical Rapid Prototyping Laboratory, National Metal and Materials Technology Center, Pathumthani, Thailand. His current research interests include biomechanics, reverse engineering and medical simulation.



Patrick Ruther (M'03) received the Ph.D. degree in mechanical engineering from the University of Karlsruhe, Karlsruhe, Germany, in 1996.

Between 1996 and 1998, he was with the Institute of Microstructure Technology IMT (Forschungszentrum Karlsruhe, Germany) as a Postdoctoral Fellow where he was responsible for industrial projects on microoptical components based on the LIGA technology. Since October 1998, he has been a Senior Scientist and Group Leader with the Microsystem Materials Laboratory, Department of Microsystems

Engineering (IMTEK), Freiburg, Germany, and a Lecturer with the Technical Faculty, University of Freiburg, Freiburg, Germany. His focus is on the design, fabrication, and characterization of CMOS-compatible microelectromechanical systems and the development of new micromachining methods for bio-medical microsystems. He was Technology Coordinator of the European research project NeuroProbes (2006–2010) developing multifunctional microelectrode arrays for cerebral applications. Since February 2013 he is General Coordinator of the EU Large Scale Integrated Project NeuroSeeker developing high-density neural probe arrays and optodes for intracortical and optogenetic applications, respectively. He is cofounder of the spin-off company ATLAS Neuroengineering, Leuven, Belgium, developing high-density silicon-based probe arrays for experimental neuroscience.



Bernd G. Lapatki received the degree in dentistry from the University of Freiburg, Freiburg, Germany, in 1994.

After working for two years as a general dentist, he defended his doctoral thesis in medicine (Dr. med. dent. degree) with the title "Quantitative Analysis for the Assessment of Functional Parameters during Trumpet Playing" in 1998. In December 2000, he finished a postgraduate study at the Department of Orthodontics, Freiburg University Medical Centre, and obtained the certificate as "specialist in orthodontics".

From September 2001 to October 2002, he attended as a Research Fellow with the Department of Clinical Neurophysiology, Radboud University Nijmegen Medical Center, The Netherlands. After finishing his habilitation thesis (leading to a German academic degree, equivalent to a Ph.D.) in dentistry with the title "Pathogenesis and treatment stability of cover-bite and Angle-Class II Division 2 malocclusion" in November 2007, he received a second Ph.D. degree from the Radboud University Nijmegen (The Netherlands) in June 2010 for his thesis "The facial musculature—characterisation at a motor unit level". In September 2009, he obtained a professorship for orthodontics from the Faculty of Medicine of the University of Ulm, where he is working as Head of the Department of Orthodontics, and since 2013 as Director of the Centre of Dentistry.

Dr. Lapatki was awarded in 1999 by the Faculties of Medicine and Applied Sciences of the Albert Ludwig University Freiburg for the best dissertation in the field of medicine/medicine technique/applied informatics/microsystems engineering. In 2004, he received the "Alex Motsch Award" of the Academy of Functional Diagnostics and Therapy (AFTD) of the German Society of Dentistry and Oral Medicine for the paper "Systematic characterisation of motor units of the facial musculature by means of high-density surface electromyography", and in 2007 the "Arnold Biber Award" (the highest scientific award

of Orthodontics in Germany) of the German Orthodontic Society for the paper “Smart brackets for 3D-force-moment measurements in orthodontic research and therapy—developmental status and prospects”.

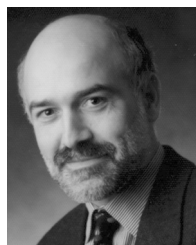


Oliver Paul (M'94) received the Diploma degree in physics and D.Sc. degree from the ETH Zurich, Zurich, Switzerland, in 1986 and 1990, respectively.

After postdoctoral work with the Fraunhofer Institute for Solar Energy Systems, he joined the Physical Electronics Laboratory at ETH Zurich in 1992 as a Lecturer and Group Leader. Since 1998, he has been a Full Professor with the University of Freiburg, Freiburg, Germany, where he heads the Laboratory for Microsystem Materials at the Department of Microsystems Engineering (IMTEK). He is a coauthor

of more than 300 technical publications, patents, and books. His group's research focuses on MEMS materials and fabrication technologies, physical microtransducers, and microstructures for industrial and life science applications. He is a member of the editorial boards of *Sensors and Actuators A: Physical* and *Journal of Micromechanics and Microengineering* and of the editorial advisory board of the *IEEE Transactions on Electrical and Electronic Engineering*. He is a cofounder of Sensirion AG and Atlas Neuroengineering. From 2006 to 2008, he served as the Director of IMTEK. He is a founding director of the German excellence cluster BrainLinks-BrainTools with the University of Freiburg.

Dr. Paul cochaired the IEEE MEMS 2004 Conference in 2004.



Yiannos Manoli (M'82–SM'08) was born in Famagusta, Cyprus, in 1954. He received the B.A. degree (*summa cum laude*) in physics and mathematics from Lawrence University, Appleton, WI, USA, in 1978, the M.S. degree in electrical engineering and computer science from the University of California, Berkeley, CA, USA, in 1980, and the Dr. Ing. degree in electrical engineering from the Gerhard Mercator University, Duisburg, Germany, in 1987.

From 1980 to 1984, he was a Research Assistant with the University of Dortmund, Dortmund, Germany, in the field of A/D and D/A converters. In 1985, he joined the newly

founded Fraunhofer Institute of Microelectronic Circuits and Systems, Duisburg, Germany, where he established a design group working on mixed-signal CMOS circuits especially for monolithic integrated sensors and application specific microcontrollers. From 1996 to 2001, he held the Chair of Microelectronics as full professor with the Department of Electrical Engineering, University of Saarland, Saarbrücken, Germany. In 2001, he joined the Department of Microsystems Engineering (IMTEK) in the Faculty of Engineering, University of Freiburg, Freiburg, Germany where he established the Chair of Microelectronics. With an endowment of the Fritz Hüttinger Foundation and in memory of the founder of today's Hüttinger Elektronik, the University of Freiburg named the Chair “Fritz Huettinger Chair of Microelectronics”, in 2010. In 2012 Professor Manoli became Dean of Engineering after having served as Associate Dean since 2008. Since 2005, he has additionally served as one of the three directors at the “Institute of Micromachining and Information Technology” of the “Hahn-Schickard Gesellschaft” (HSG-IMIT) in Villingen-Schwenningen, Germany. His current research interests are the design of low-voltage/low-power mixed-signal CMOS circuits, energy harvesting electronics, sensor read-out circuits as well as Analog-to-Digital converters. Additional research activities concentrate on motion and vibration energy transducers and on the field of inertial sensors and sensor fusion. In 2000, he had the opportunity to spend half a year on a research project with Motorola (now Freescale) in Phoenix, AZ, USA. In 2006, he spent his sabbatical semester with Intel, Santa Clara, CA, working on the read-out electronics for a high-resolution accelerometer.

Prof. Manoli and his group have received best paper awards at ESSCIRC 1988, 2009 and 2012, PowerMEMS 2006, MWSCAS 2007, and MSE-2007. The MSE-2007 award was granted for SpicyVOLTSim (www.imtek.de/svs), a web-based application for the animation and visualization of analog circuits for which Yiannos Manoli also received the Media Prize of the University of Freiburg in 2005. He was the first to receive the Best Teaching Award of the Faculty of Engineering when it was introduced in 2008. For his creative and effective contributions to the teaching of microelectronics, he has also received the Excellence in Teaching Award of the University of Freiburg and the Teaching Award of the State of Baden-Württemberg, both in 2010. He has served as a Distinguished Lecturer of the IEEE and as guest editor of the IEEE TRANSACTIONS ON VERY LARGE-SCALE INTEGRATION (VLSI) SYSTEMS and the IEEE JOURNAL OF SOLID-STATE CIRCUITS. He is on the Senior Editorial Board of the IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS and on the Editorial Board of the *Journal of Low Power Electronics*. He has served on the committees of a number of conferences such as ISSCC, ESSCIRC, IEDM and ICCD, and was Program Chair (2001) and General Chair (2002) of the IEEE International Conference on Computer Design (ICCD). He is a member of VDE, Phi Beta Kappa, Mortar Board, and senior member of the IEEE.